

Model	Params	FP16 VRAM	Q4 VRAM	RTX 4090 (24GB)	RTX 6000 Ada (48GB)	H100 80GB	GPUs for 1000 Images/Min	GPUs for 10000 Images/Min	Recommended Use Case
Qwen2.5-VL-7B	7B	~14 GB	~4-5 GB	1 GPU	1 GPU	1 GPU	H100: 1 • RTX6000: 1 • RTX4090: 1	H100: 2 • RTX6000: 3 • RTX4090: 4	Cost-sensitive deployments
Gemma 3 12B	12B	~24 GB	~7 GB	1 GPU	1 GPU	1 GPU	H100: 1 • RTX6000: 1 • RTX4090: 1	H100: 2 • RTX6000: 3 • RTX4090: 4	Balanced enterprise entry
Gemma 3 27B	27B	~54 GB	~16 GB	4 GPUs FP16 / 1 GPU Q4	2 GPUs FP16 / 1 GPU Q4	1 GPU	H100: 1 • RTX6000: 1 • RTX4090: 1(Q4)	H100: 2-3 • RTX6000: 4-5 • RTX4090: 6-8	Enterprise deployment
Qwen2.5-VL-32B	32B	~64 GB	~16-20 GB	4 GPUs FP16 / 1 GPU Q4	2 GPUs FP16 / 1 GPU Q4	1 GPU	H100: 1 • RTX6000: 1 • RTX4090: 1(Q4)	H100: 3-4 • RTX6000: 5-6 • RTX4090: 8-10	Best ROI for production OCR
Qwen2.5-VL-72B	72B	~145 GB	~36-48 GB	8 GPUs FP16 / 2-3 GPUs Q4	4 GPUs FP16 / 1-2 GPUs Q4	2-3 GPUs FP16 / 1 GPU Q4	H100: 1(Q4) • RTX6000: 2 • RTX4090: 3	H100: 6-8 • RTX6000: 10-12 • RTX4090: 16-20	Maximum accuracy